Analyzing Key Influential Factors for Achieving 5-Point Math Entitlement in Israel

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1. Introduction

Our research question is: How do different social, financial, and educational elements affect the entitlement for 5 points Bagrut in mathematics in Israel?

This is an important question because it can help us understand the factors contributing to students' success in mathematics. This information can then be used to develop interventions that can help to improve students' mathematical achievement.

The general problem area that this analysis contributes to is the field of educational research. Educational research is the study of how students learn and how to improve educational outcomes. This research can be used to ensure that all students have the opportunity to succeed.

This is a hard question to answer as the data required to address it is located in many different datasets with numerous relevant features, making it difficult both to collect the data and to properly analyze it.

Although prior work on this specific research question is limited, we have found several additional research studies that provide relevant information. Firstly, we have found a report by the state comptroller from 2016 that highlights a decline in the percentage of students achieving 5 points in math Bagrut in recent years. This report also includes graphs illustrating the influence of sector and socioeconomic status on the rate of change in the entitlement for 5 points Bagrut. Moreover, The Information and Research Center of the Knesset has conducted a few pieces of research on the subject. One of the studies, for example, examines the entitlement for 5 points Bagrut in mathematics from various perspectives. It analyzes the entitlement by socioeconomic status, nurturing index of the Ministry of Education, and geographical district. The other report focuses on the changes in the budget per student in the education system in recent years. Additionally, we have come across news articles published annually that provide information on the percentage of students achieving the 5 points math Bagrut per school in Israel.

Despite these existing sources, our research takes a unique approach by combining several sources of data on schools and by analyzing parameters that have yet to be properly assessed. Utilizing this comprehensive approach, we will use statistical methods to identify the most significant predictors that impact students' entitlement for 5 points Bagrut in mathematics.

2. Data Overview

The dataset contains various tables with information about high schools, municipalities, socio-economic factors, and educational measures in Israel. It includes data on school budgets, socio-economic status, teacher-student relations, Bagrut results, and more. There are three entities in our dataset: Schools, Municipalities, and Socioeconomic groups. Provided below is a list of family features appropriate for tackling the research question:

- Financial Features: These include variables related to high school budgets, such as teaching hours, teaching costs, tutor hours, tuition fees, and total budget allocations.

- Socio-Economic Features: These encompass variables related to the socio-economic status of municipalities and districts, including socio-economic status and nurturing index (pentile).

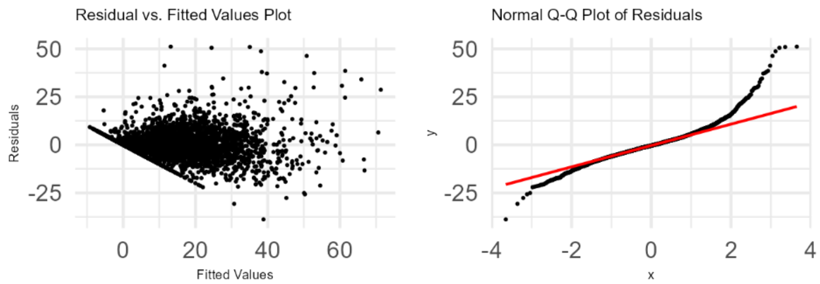
- Educational Features: These features capture various aspects of the educational environment. They include variables related to teacher-student relations, teacher satisfaction, violence, interest, teacher development, and teaching methods.

- School-Specific Features: These include variables related to individual schools, such as school type, sector, and specific school-level characteristics. They capture the unique attributes of each school.

3. Methods and Results

We chose to use the linear regression model for modeling our database. Linear regression allows us to analyze the relationship between the dependent variable (entitlement for 5 points Bagrut in mathematics for regular schools, hereafter entitlement) and multiple independent variables (social, financial, and educational elements). It helps quantify the strength and direction of these relationships, providing insights into how changes in the independent variables influence the dependent variable. This approach is suitable for our research question as we are seeking to find which features are influential on entitlement. Our model has an adjusted R-squared of 0.7303 and an RSE of 7.229.

Before we could conclude that the model we used does in fact fit the data, we conducted 3 tests to check the linearity and homoscedasticity.



Residuals vs Fitted: In this graph, we found that the residuals are spread around a horizontal line without distinct patterns - a good indication of having linear relationships.

Normal Q-Q: As displayed in the above graph, the residuals are lined well on the straight dashed line meaning the residuals are normally distributed.

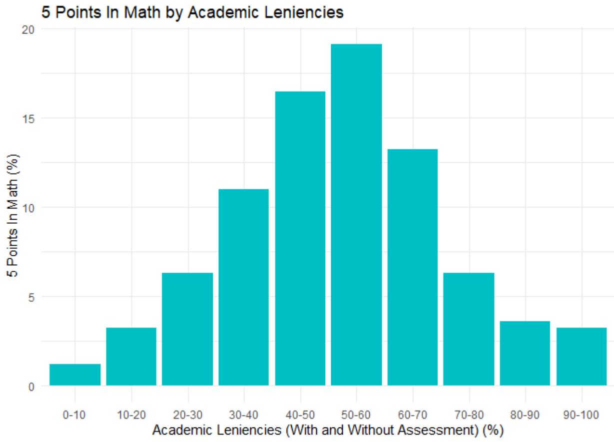
Using this model, we've found some key fields that are strongly correlated to the target variable. For example, we have found:

- Schools where teachers have a higher rate of master's degrees, have higher entitlement.
- Schools with a higher academic leniencies rate have a higher entitlement.
- Schools of a lower pentile (higher nurturing index) have higher entitlement.
- Schools with a higher special-ed score significantly less in entitlement
- The sector largely influences entitlement, the direction of which depends on the sector.

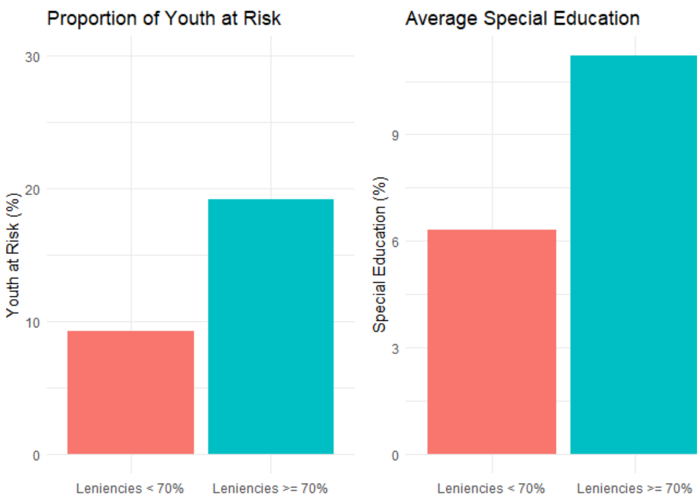
Modeling the data with linear regression helped us better understand the dataset and enabled us to focus on the more significant variables with further analysis. Moving forward, we handpicked some of the aforementioned features and others seeking to explain results from the model and other key relationships in the data. Insights regarding key features of entitlement relationship in this space are presented below:

1) The Academic Leniencies – Entitlement relationship

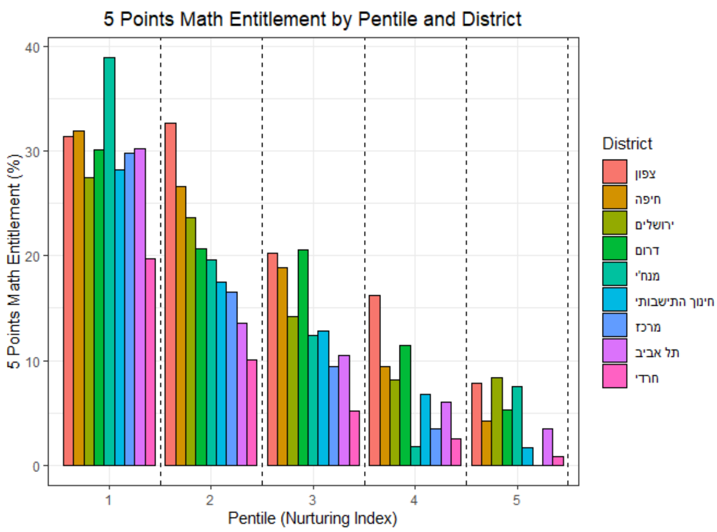
When creating a bar chart of this relationship we came across an interesting phenomenon in which there is a positive relationship between academic leniencies and entitlement until a high point in the 0.5-0.6 bar where afterward there is a negative relationship between the two. We further investigated this feature to understand potential factors that could explain this. We checked the possibility that the positive relationship could be explained by schools with similar backgrounds, meaning that the rise in entitlement could be directly influenced by the higher academic leniencies rate in that school. Meanwhile, the negative correlation could be explained by a different factor – schools with an extremely high academic leniencies rate might be of a different type and may represent a higher percentage of special schools for either students with bad grades or learning disabilities. We checked this with the following graphs.



To explain the rise in the first half of the bar chart, we took a group of the same characteristics and checked for correlation between entitlement for 5 points in math and academic leniencies. To our surprise, there was near to no correlation between the two elements in the graph. Meaning that there must be a different reason for this positive correlation (mentioned in section 4).



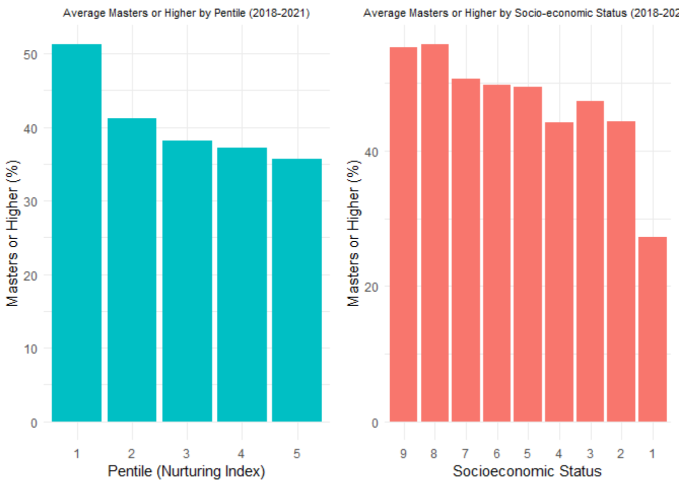
In the following graph, we searched to explain the negative correlation by checking if schools with higher academic leniencies rates have a different background than schools with lower academic leniencies rates. And in fact, we found that schools with higher than 70% academic leniencies included around twice as many schools for youth at risk and special education.



Moving forward, we assessed the relationship between the district and the entitlement over each pentile[1]. There is a clear and unsurprising correlation between the pentile and entitlement, but the magnitude of which is unconventionally dramatic. When dividing the districts per pentile we found out how the northern district overperforms while the ultra-orthodox performs badly (mentioned in section 4).

[1] Pentile – Nurturing index per school by the Ministry of Education. Consists of: 40% level of education of most educated parent, 20% family income per person, 20% school's level of outlying areas, and 20% combination of immigration and underdeveloped countries.

2) Following the findings from our linear regression model in which there is a high positive correlation between teachers' master's or higher degree and student entitlement, the following graph depicts the strong connection between the pentile and socioeconomic status and the master's or higher degree. This connection shows that well-educated teachers can be found in stronger social classes, a truth that in turn may largely impact the students' entitlement.



4. <u>Limitations and Future work:</u>

One limitation we have is that the school size is not represented in our findings and had we had a larger timeframe we would've taken into account schools and cities/municipalizes as weighted elements. Secondly, the database we created is limited to a timeframe of approximately 6 years and this research question requires a larger timeframe for both more data records and for a better understanding of tias me a factor. Thirdly, acknowledging that correlation does not necessarily imply causation, it is very difficult to answer the question of which factors are influential as it implies causation. Lastly, this is a very complex subject with numerous other factors that have not been included in the database.

Given more time, we would also conduct further analysis to answer unsolved questions. We would seek to find the reason for the positive correlation in the academic leniencies – entitlement relationship as aforementioned. Furthermore, we would investigate the reason for the drastic difference in the different districts' entitlement and search for key factors behind this gap. In addition, we would dive into the budget and its distribution, trying to understand the state today and how it could be effectively changed for the better.

Our code repository on Github:

https://github.com/Fabian665/Final-Project-R

Link to our data folder

https://drive.google.com/file/d/10v8pYjHERq00c8_sIv2WJFzAXZVk2cqy/view?usp=drive_link

Appendix:

All data collected is published on the Ministry of Education website:

https://infocenter.education.gov.il/all/extensions/Shkifut_Reports/Shkifut_Reports.html